

Vol 9 Chapter 7 — Phonons

Keeper (author pass — deep math/physics revision)

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Chapter 7 — Phonons

Level 1 — one sentence

Phonons are quantized lattice vibrations — substrate vibrational K-types of the periodic ionic configuration — with acoustic branches (linear dispersion at low $\vec{k}$, governing sound and elasticity) and optical branches (gapped at $\vec{k} = 0$, governing IR absorption), and they mediate the BCS Cooper-pair attraction for conventional superconductors (Ch 4).

Level 2 — graduate-physicist precision

7.1 Lattice vibrations

For a 1D chain of identical atoms with nearest-neighbor spring constant K and lattice spacing a :

$$\omega(k) = 2\sqrt{K/M} |\sin(ka/2)|$$

Linear at small k (sound waves with speed $v_s = a\sqrt{K/M}$); maximum at zone boundary $k = \pi/a$.

For diatomic chain (two atoms per unit cell): two branches — acoustic (gapless at $k = 0$) and optical (gapped at $k = 0$).

In 3D: 3 acoustic branches (longitudinal + 2 transverse) + $(3N-3)$ optical branches for N atoms per unit cell.

7.2 Quantization: phonons

Each normal mode is a quantum harmonic oscillator (Vol 5 Ch 4). Quantum excitations are phonons with energy $\hbar\omega$, momentum $\hbar k$. Bosons (integer spin).

Total energy of crystal: $E = \sum_{n,k} \hbar\omega_n(\vec{k}) [n_{nk} + 1/2]$ with n_{nk} occupation numbers (Bose-Einstein distributed at temperature T).

7.3 Debye and Einstein models

Einstein model (1907): all phonons at single frequency ω_E . Specific heat $C_V = 3R(\hbar\omega_E/k_B T)^2 e^{\hbar\omega_E/k_B T} / (e^{\hbar\omega_E/k_B T} - 1)^2$. Reduces to $3R$ at high T (Dulong-Petit); drops as $e^{-\hbar\omega_E/k_B T}$ at low T .

Debye model (1912): linear acoustic dispersion up to Debye cutoff ω_D , $k_B T_D = \hbar\omega_D$. Gives: - Low T : $C_V \propto T^3$ (matches experiment for non-metals) - High T : $C_V \rightarrow 3R$ (Dulong-Petit limit)

Debye temperatures: Pb ~ 88 K (soft), diamond ~ 2230 K (stiff).

7.4 Phonon-electron coupling

Phonons scatter electrons $\rightarrow$ resistivity in metals ($\rho \propto T$ above Debye T; $\rho \propto T^5$ at low T from acoustic phonons).

Phonons mediate Cooper-pair attraction in BCS superconductors (Ch 4): two electrons exchange a virtual phonon $\rightarrow$ effective attraction overcoming Coulomb repulsion below T_c .

7.5 Anharmonic effects

Beyond harmonic approximation: thermal expansion, phonon-phonon scattering (thermal conductivity at high T), Grüneisen parameter, Mössbauer effect — all anharmonic phenomena.

7.6 K-audit anchors

- Vol 5 Ch 4: quantum harmonic oscillator (single phonon mode)
- Vol 6 Ch 7: Debye model in stat mech
- Vol 8 Ch 9: continuum elasticity (long-wavelength phonons)
- Ch 4 of this volume: phonon-mediated BCS superconductivity

Level 3 — 5th-grader accessibility

Phonons are quantized lattice vibrations — sound waves in crystals come in discrete quanta, just like light comes in photons. Two types of branches: - Acoustic phonons: long-wavelength = sound waves; gapless at $k = 0$ - Optical phonons: high-frequency vibrations of unit cell; gapped at $k = 0$ (absorb IR light)

Debye model: gives specific heat $C_V \propto T^3$ at low T (matches experiment); reduces to Dulong-Petit $3R$ at high T. Phonons scatter electrons (causing resistivity). Phonons mediate Cooper pairing in conventional superconductors (BCS).

What comes next

Chapter 8 develops the BaTiO₃ 137-plane substrate eigentone — BST's signature condensed-matter falsifier.

Where to look this up

- Ashcroft and Mermin Ch 22-24
- BST: Ch 4 (phonon-BCS link); Vol 5 Ch 4